

12

20



Bag A



Bag B

Bag A contains 5 white balls and 3 black balls.

Bag B contains 3 white balls and 1 black ball.

- (a) Two balls are picked at random from bag B without replacement.

Find the probability that both balls are black.

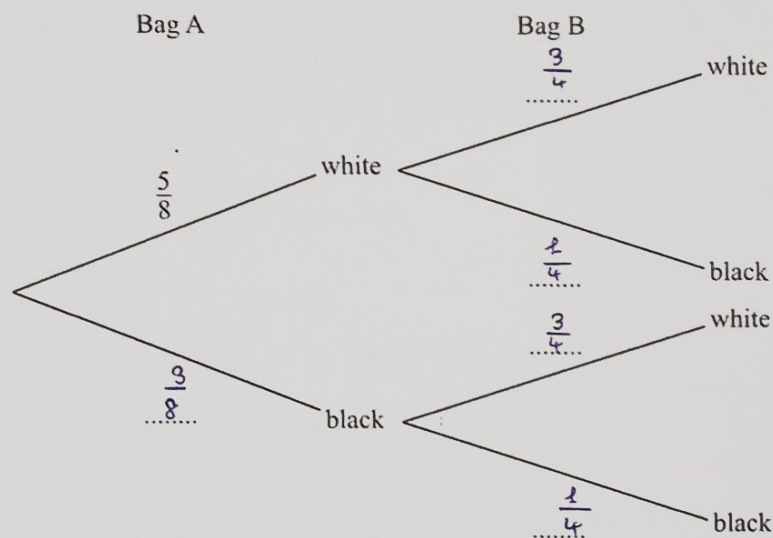
$$\frac{1}{4} \times \frac{1}{3} = \frac{1}{4} \times \frac{0}{3}$$

$$= \frac{0}{12} = 0$$

..... [1]

- (b) The balls are replaced into bag B.
Kyle picks a ball at random from each bag.

- (i) Complete the tree diagram.



[2]

- (ii) Find the probability that the two balls are the same colour.

$$= \left(\frac{5}{8} \times \frac{3}{4} \right) + \left(\frac{3}{8} \times \frac{1}{4} \right)$$

$$= \frac{15}{32} + \frac{3}{32} = \frac{18}{32} = \frac{9}{16}$$

..... [3]



- (c) The balls are replaced into their bags.

Jo picks a ball at random from bag A and places it into bag B.
She then picks a ball at random from bag B.

Find the probability that she picks a black ball from bag B.

bag A black

$$= \frac{3}{8}$$

bag B black

$$= \frac{2}{4}$$

New black total

$$= \frac{3}{8} + \frac{2}{4} \times \frac{1}{2}$$

$$= \frac{5}{8}$$

$$\frac{5}{8}$$

[3]